

Crime Track - Online Crime Reporting System

Category: Full Stack Development

Skills Required: Node.js, MongoDB

1. Overview:

Crime Track - The Crime Reporting Platform aims to provide a comprehensive and user-friendly solution for individuals seeking to report criminal activities while also accessing vital information about various crimes. The platform allows users to report incidents in their area with details including type of crime, location, and relevant evidence. Additionally, it provides educational resources, crime statistics, and community discussions for awareness and safety.

2. Scope:

In Scope:

- User Registration and Login
- Crime Reporting Form
- Crime Information Dashboard with prevention tips and legal actions
- Community discussion forum and real-time alerts

Out of Scope:

- Law enforcement tools for crime analysis
- Crime solving assistance

3. Requirements:

Functional Requirements:

- User authentication (registration, login, JWT token)
- Crime report submission with details (type, location, time, evidence)

- Information dashboard for crime types and prevention tips
- Notification system for crime alerts
- API to fetch crime reports, search by location, and assign to officers

Non-Functional Requirements:

- Security of user data (encryption of passwords, secure authentication)
- Scalability for handling multiple crime reports and user data
- Performance optimization for real-time crime data updates

4. Architecture and Design:

System Architecture:

- Backend: Node.js with Express.js
- Database: MongoDB for storing user data and crime reports
- Frontend: React for building an intuitive interface (optional future enhancement)

Database Design:

- User Collection: Stores user details (ID, Name, Email, Age, Phone, Crime Reports)
- Crime Reports Collection: Stores crime details (Type, Location, Date, Time, Evidence)

Authentication:

- JWT (JSON Web Token) based user authentication system.

ER Diagram:

- User: UserID, Name, Email, Phone, Report, Age
- Report: ReportID, Crime Type, Location, Evidence, Status

5. Key Features:

1. User Dashboard: Displays types of crimes, prevention tips, and a crime reporting form.
2. Crime Report Form: Allows users to report any suspicious activity or crime.
3. Crime Information Section: Lists various types of crimes and their legal consequences.
4. Real-time Alerts: Notifications and alerts based on user subscriptions.

6. Project Setup and Configuration:

1. Install required tools:

- Node.js and npm
- MongoDB (local or cloud-based)
- Express.js

2. Create project folders:

- Backend: Includes server setup, API routes, models, middleware

3. Install Backend Packages:

- Express.js, dotenv, Mongoose, JWT, Cors, Body-parser

7. Backend Development:

1. Setup express server:

- Create index.js file for server setup.
- Configure routes for user registration, login, crime report, and profile.

2. User Authentication:

- Implement JWT-based authentication for login and signup.

3. API Routes:

- /api/users/register: User registration.
- /api/users/login: User login and JWT token generation.
- /api/reports: Crime reporting API (POST method).
- /api/reports/search: Search crime reports by location.
- /api/reports/:id/status: Update crime report status.

8. API Testing:

1. Postman or ThunderClient can be used to test the API endpoints.
2. Ensure the following API functionalities are working:
 - User registration, login, and JWT generation.
 - Crime report creation, search, and status update.